SPE-229898-MS

Improving Injection Efficiency with a Novel Autonomous Injection Valve

T. T. Larsen, H. A. Dsouza, Henrik Ågrav, and I. M. Ismael, Inflow Control AS, Norway

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

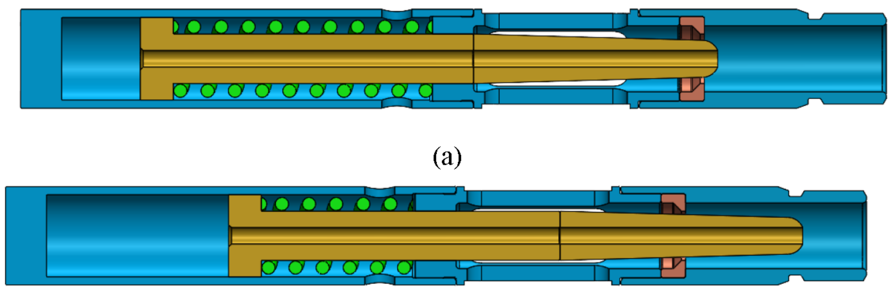
Meeting the world's growing energy demand requires improved recovery from existing oil reservoirs. Conventional injection methods often suffer from fluid loss into high-permeability zones, leading to early water breakthrough and reduced sweep efficiency. This paper introduces a novel autonomous injection valve designed to deliver a constant volumetric flow rate independent of differential pressure.The valve leverages fluid dynamic principles to autonomously regulate flow without external control. A comprehensive qualification program was conducted, including flow performance testing, mechanical integrity assessments, and evaluations in erosive and completion fluid environments. Additionally, a reservoir simulation model was developed to benchmark the valve against conventional injection technologies. Results demonstrate that the injection valve maintains a stable flow profile across a wide pressure range, with tunable flow rates derived from its mathematical definition. Simulation outcomes indicate that uniform injection enabled by this valve reduces fluid losses, enhances sweep-efficiency, and increases recovery—while also lowering injection capacity requirements and associated carbon emissions. This technology offers a step-change in injection control, enabling more efficient reservoir management and reduced environmental impact. Its predictable flow behavior and autonomous operation present a compelling alternative to conventional injection strategies.

Introduction

As global energy demand continues to rise, the oil and gas industry is under increasing pressure to maximize recovery from existing assets. One of the key challenges in injection-based enhanced oil recovery (EOR) methods is the uneven distribution of injected fluids, particularly in heterogeneous or fractured reservoirs. High-permeability streaks or fractures often dominate the flow path, leading to early water breakthrough and poor sweep efficiency. To address this, a new autonomous injection valve has been developed. Unlike conventional passive injection devices, this valve dynamically adjusts to changing pressure conditions to maintain a constant flow rate, thereby improving the uniformity of injection and enhancing overall recovery.

Working principle

The valve utilizes the hydrodynamic forces exerted by the moving injection fluid to change the flow opening autonomously to achieve a constant flow rate over a range of differential pressures. To accomplish this, the geometry and the components of the valve are carefully defined mathematically with a force balance of the valve system including contributions from fluid mechanical- and mechanical forces in the system. A section view of the valve is shown in Fig. 1. The movement of the valve internals are therefore defined by the geometry, the actual flow rate and the differential pressure over the valve. As differential pressure increases, the valve internals move to reduce the flow area to counteract the natural tendency of the flow to increase (as it would in a passive device).



(a)

(b)

Figure 1—Section view of the main components in the valve. Showing the moving member (gold), the compliant member (green) and the choking area (brown). The figure shows the valve in (a) Fully open position (b) Fully choked position. As the moving member moves to the right the effective flow area reduces.

Internal flow geometry of the valve is determined by a mathematical model where target flow rate is one of the inputs. Fig. 2 shows an example output from the model. The actual geometry of the valve is anonymized in this example, but the example is made to show how the shape of the moving element changes its geometry based on the target flow rate.

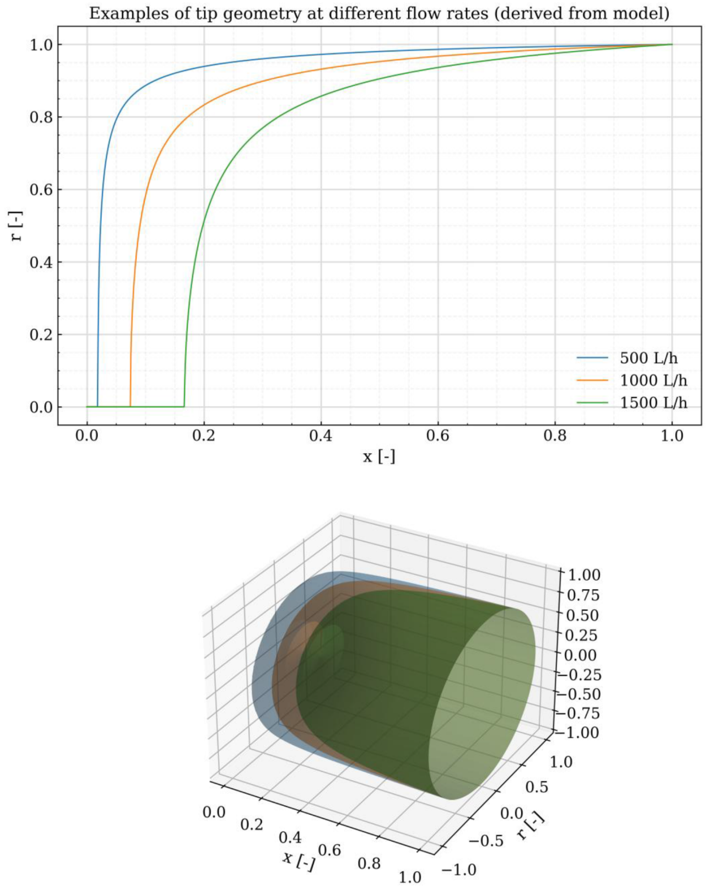


Figure 2—Anonymized output from mathematical model that generates the geometry of the valve internals. The example shows how geometry changes with target flow rate. The upper plot shows the profile of the moving member while the lower plot shows a revolved 3d-projection.

The valve has three stages of operation, described in Fig. 3 below.

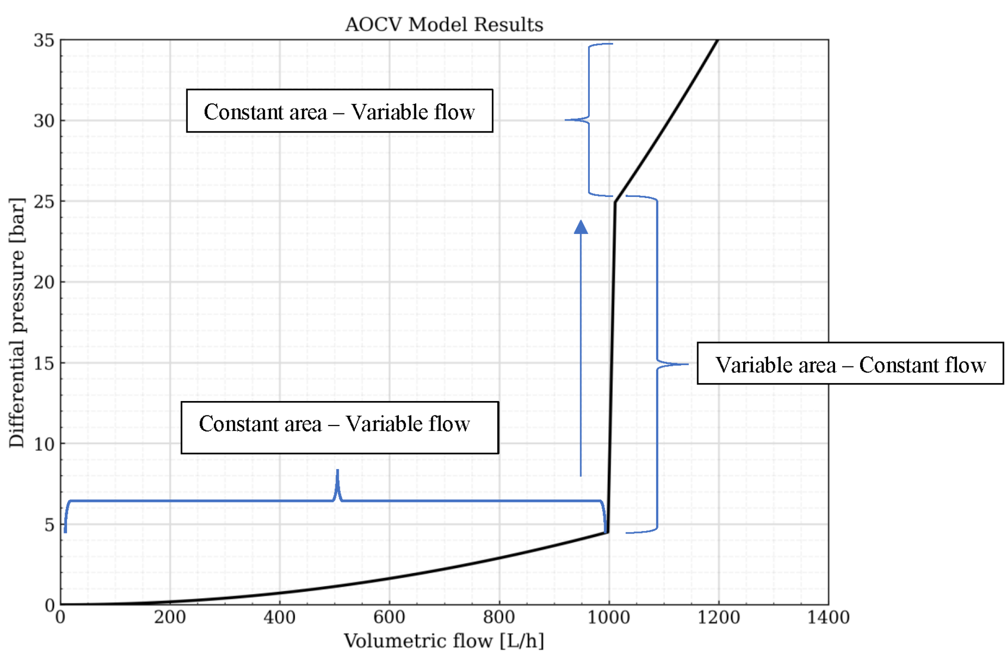


Figure 3—Illustration showing the three operating areas of the valve.

Initially the valve behaves like a conventional ICD with constant flow area until it reaches the goal flow rate. At the goal rate the compliant member and the flow forces are in perfect equilibrium. As the differential pressure is further increased, the valve moves into the constant flow area where the valve balances the natural tendency of flow increase due to increased differential pressure, by decreasing the flow are aautonomously. At a given maximum differential pressure, the compliant member reaches its maximum stroke, and the valve behaves like a constant-area ICD again. This process is fully reversible. For further technical explanation it is referred to [1].

Test setup and program

Table 1—Summary of qualification tests performed during the development.

Flow performance test	0-20 bar differential pressure water at approximately 20°C
Mud test/flow initiation test	Flowrate at around 7 bar differential pressure Pre hold– Post 4 hours hold- Post 17 hours hold
Erosion test	2 ppm over 10 years 20 bar differential pressure 1000 L/h 20°C accelerated over 110 hours.

Test results

Performance test

Results from the performance test of the valve with fresh water at 20°Care shown in Fig. 4.

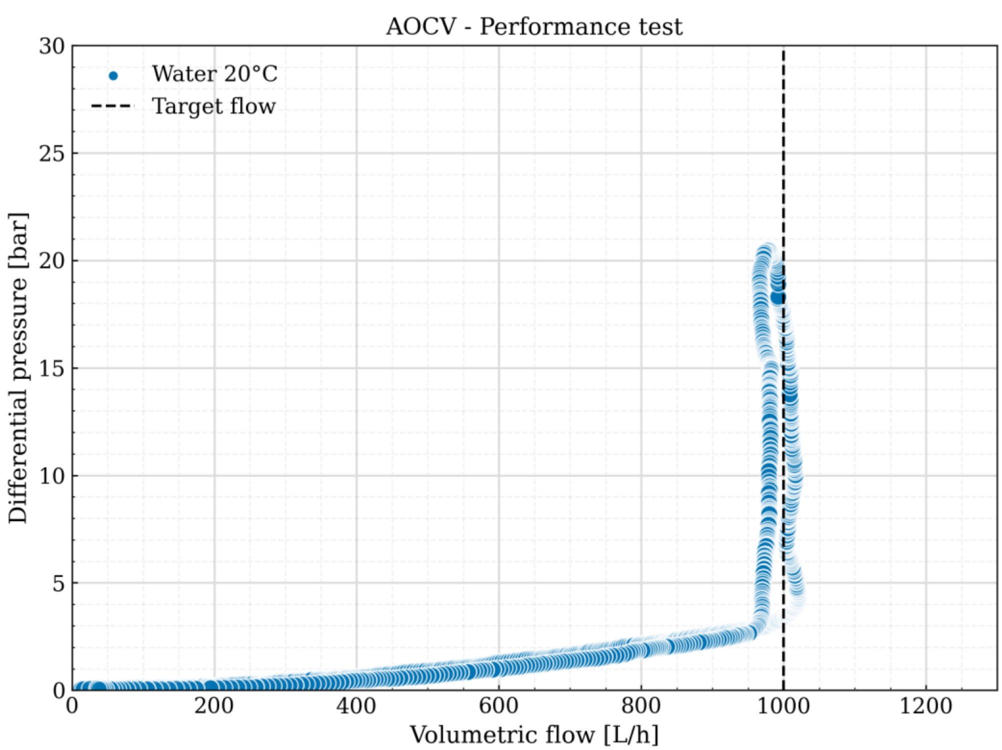


Figure 4—Flow performance test of the injection valve with water at 20°C, showing the constant flow behavior after the goal rate of 1000L/h is reached.

The valve achieved the target flow rate of 1000 L/h at a differential pressure of 2.5 bar and maintained constant flow up to the maximum test pressure. The observed hysteresis loop, attributed to internal valve friction, was anticipated in the design. The right leg of the loop corresponds to increasing flow rates, the left to decreasing rates. Valve performance met expectations. Testing at higher differential pressures was limited by the rig's cavitation threshold.

Mud and flow initiation test

The test was conducted in a dedicated mud test rig in the normal flow direction. The objective was to determine whether the technology would clog after a designated settling period. The test matrix involved a sediment plugging test, allowing the test fluid to rest for 4 hours and 17 hours respectively. Before and after each settling period, a flow initiation test was conducted and flow rate at a set reference point with mud at around 7 bar differential pressure was measured using a Coriolis flow meter.

The test results showed consistent flow performance after the 4-hour and 17- hour settling time. The flow initiation test indicated no sign of plugging. The flow rates measured at around the reference point were at all instances around 900 kg/h.

Results showing the reference point at different hold times are given in Fig. 5.

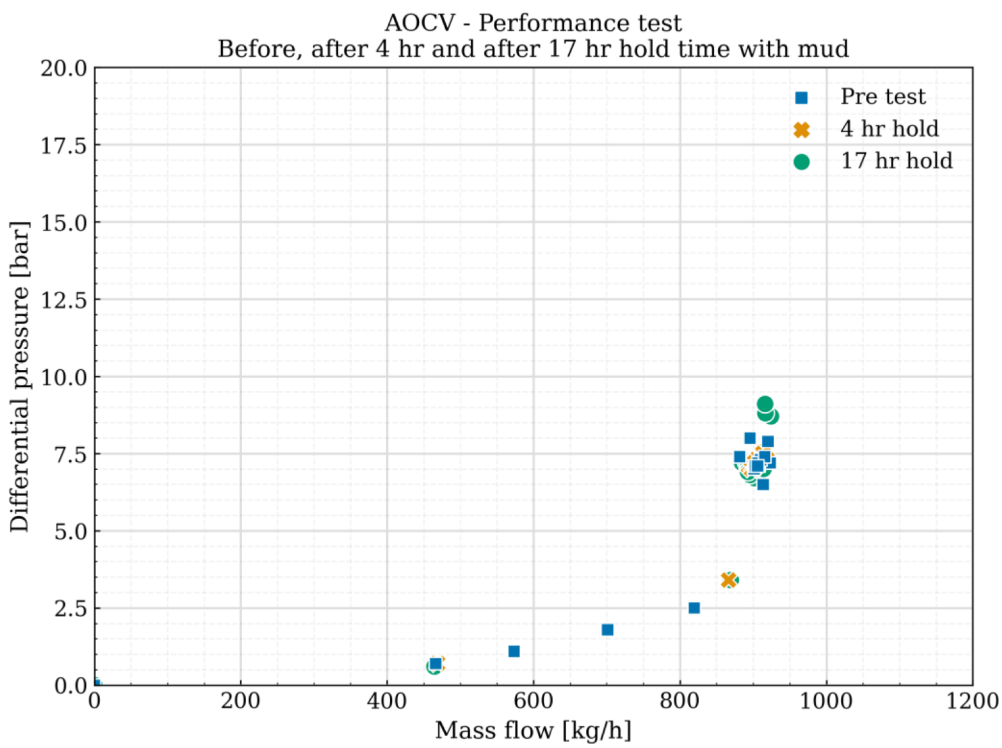


Figure 5—Performance before and after submerging the valve in mud. No changes in performance were observed.

Erosion test

An erosion test on the valve was conducted to test the durability of the valve parts as a part of the general qualification program. The erosion test was carried out with silica sand and water as carrier fluid. It was conducted to simulate a constant sand load of 2 ppm over 10 years. The sand concentration was kept at an average of 2000 ppm over a duration of 110 hours. Performance curves for the valve before, during and after the test are shown in Fig. 6.

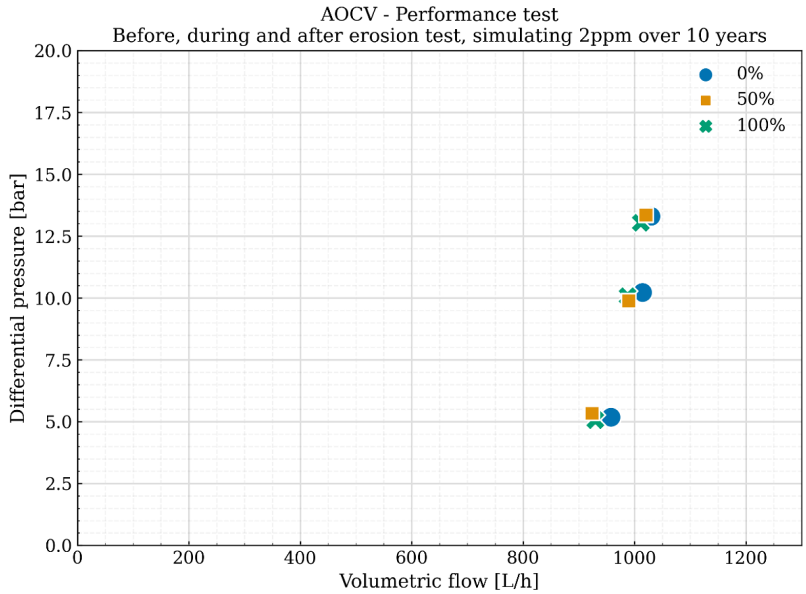


Figure 6—Performance of the valve during under and after the erosion test showing negligible changes in performance. The percentages define percentage completion of the test where 100% is 2 ppm over 10 years. Points show average values over time at discrete differential pressures.

The results from erosion testing indicate that the valve exhibited no erosion damage that impairs performance throughout the test duration. The flow rates measured prior to the test, as well as at 50% and 100% completion, were consistent at specified discrete differential pressures. All data points remained within the established hysteresis loop. Time series of test data averaging over a period of 60 min are shown in Fig. 7 showing no significant changes in performance over time.

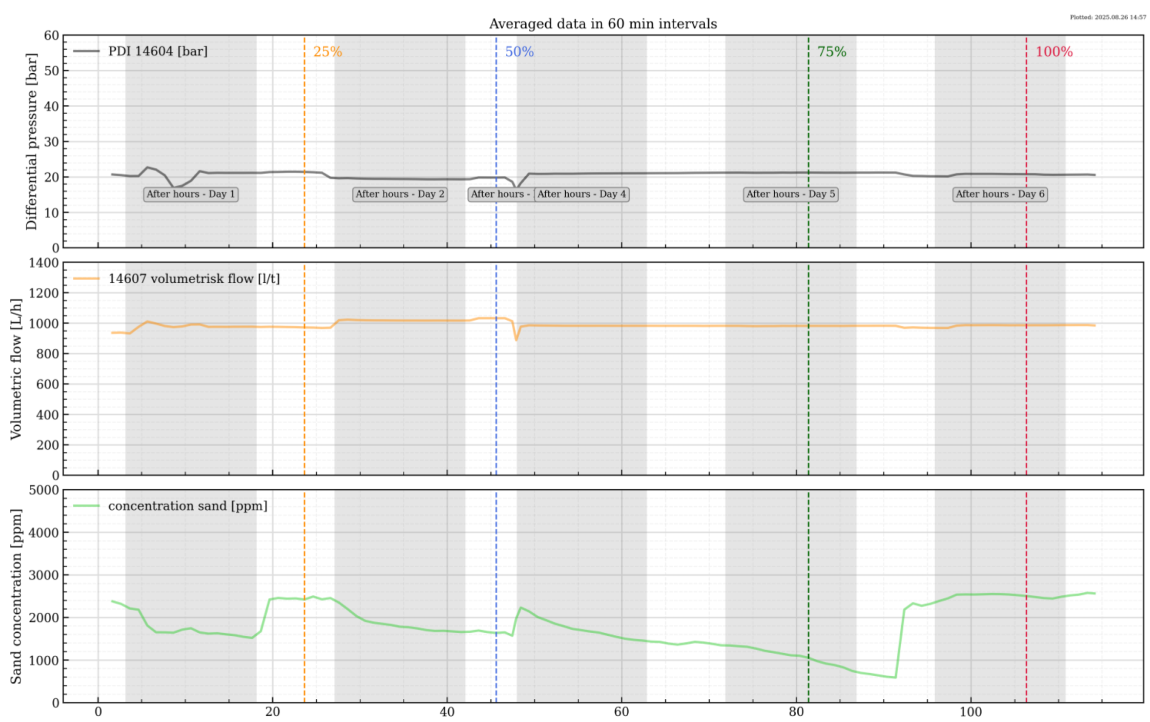


Figure 7—Main parameters during test. Inspection and performance tests at 50 and 100% completion (blue and red vertical dashed line).

Static modelling with autonomous injectors

To highlight the effectiveness of the autonomous injector, a static model has been set up for a horizontal injector well. The simulator used is an industry standard static steady-state near wellbore simulator. The injector consists of a long injector of 2500 m. The formation is in a sandstone reservoir that is prone to fractures. Since fractures are difficult to predict and manage, two sets of permeability profiles along the wellbore have been used for simulation work. The injector-well will target an injection rate of 1900 Sm3/d. For comparative purposes three cases have been set up:

Case A: OH injector

Case B: Passive ICD injector

Case C: Autonomous injector

The first permeability profile is based on a hydrodynamic model with several high permeable streaks along the wellbore with a low permeable section in the toe. The second permeability profile is similar to the original one but has been updated with several high permeable fractures close to the toe. Both permeability profiles can be seen in Fig. 8.

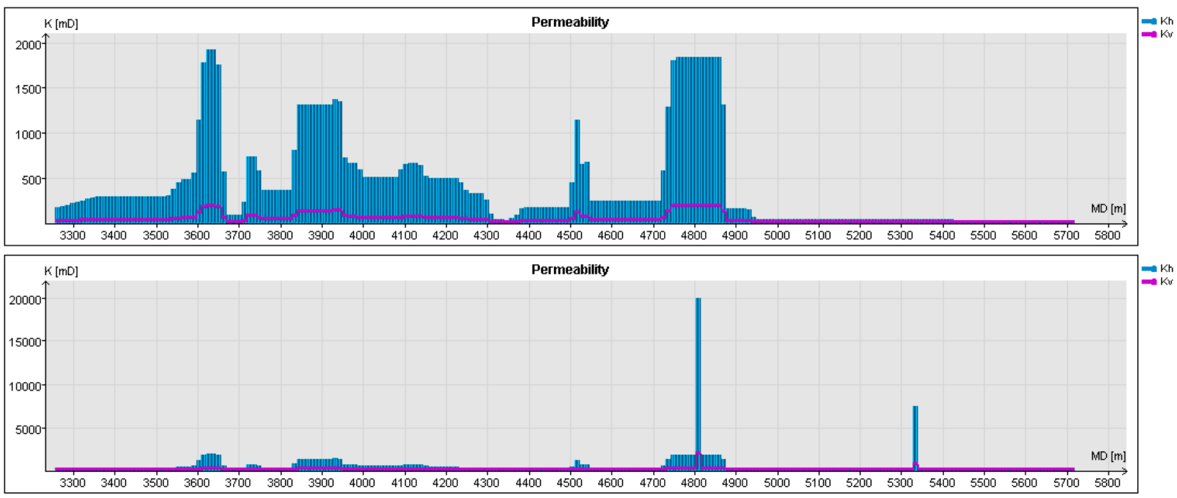


Figure 8—Permeability profiles along the wellbore. Top shows initial unchanged profile, while bottom shows modified permeability with added fractures.

Static simulations have been performed for all three cases with the same injection rate of 1 900 Sm3/d, and all cases have been simulated with both sets of permeability profiles. For Case A the well as been modelled with no lower completion. For the other two cases the well has been segmented into 10 compartments separated with packers. The number of autonomous injectors has been decided to match the expected flow rate of 1 900 Sm3/d. Sizing of ICDs has been done to match the performance of the autonomous injector prior to the constant rate section that occurs after 2.5 bars.

A comparison of the pressure drops across completion for the autonomous injector and ICD completion for the first permeability profile can be seen in Fig. 9, as well as a comparison of the cumulative injection along the wellbore. For the differential pressure across the autonomous injection valvesit is fairly high in the higher permeability section of the wellbore, staying at around 30 bars with some zones reaching higher differential pressure than 30 bars. This means that the autonomous injector completion is restricting flow in these sections to the constant rate of 25 m3/d, with some section injecting slightly more. This indicated that the autonomous injector is working as it is supposed to, restricting unnecessary injection of water into the high permeable sections by applying higher differential pressure in these compartments. For the low permeability section, the DP across the autonomous injectors is relatively low to allow as much flow as possible into the low permeable toe section for this well. The DP across the ICD section is lower but still following the same general trend along the wellbore as the autonomous injector. The cumulative injection comparison indicates that the autonomous injectoris providing better injection in the low permeable toe section compared to the ICD injector. In the last three zones in the toe the AOCV is injecting 400 Sm3/d which is double the injection of the ICD completion which is at 200 Sm^3/d.

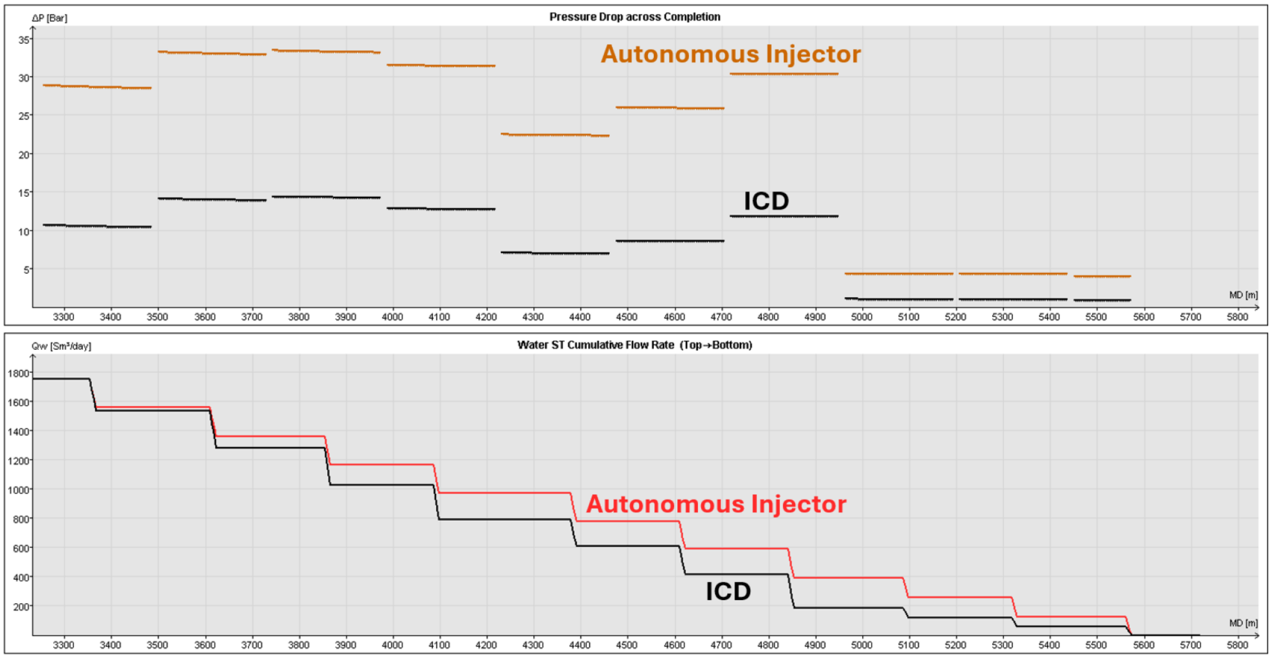


Figure 9—Downhole pressure profile and cumulative production for the first permeability profile. Top: Downhole differential pressure across autonomous injector and ICD completion. Bottom: Cumulative injection along completion for autonomous injector and ICD completions

The difference is further highlighted in Fig. 10 which shows a bar chart for the flow distribution for Case A (dotted), B (striped), and C (full color) using the first permeability profile. Case A with the OH well injects mostly into the high permeable sections with almost no water being injected into the toe section. Case B with ICD completion does a decent job at balancing the injection, but still struggles to inject into the toe section. Case C with the autonomous injection has an even injection all along the wellbore. There is a bit less injected into the toe compared to the other compartments, about 8% of the total flow in each of the 3 toe compartments.

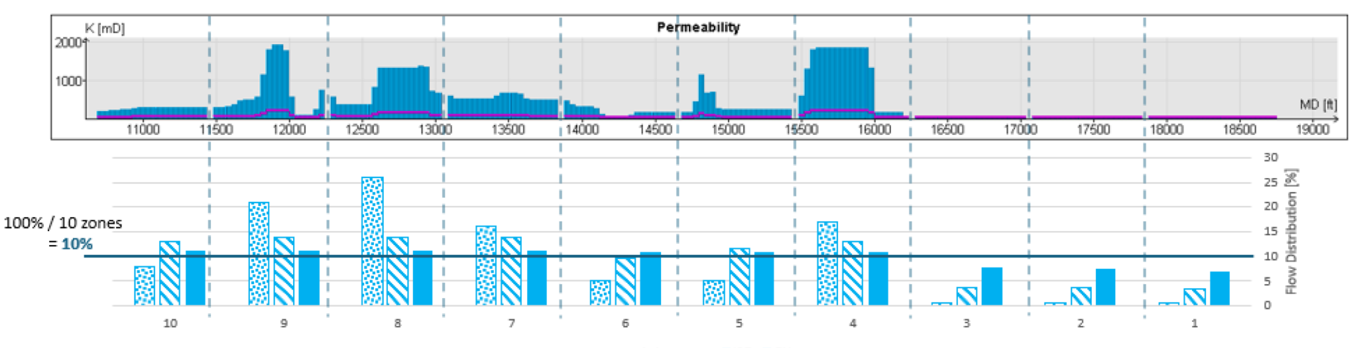


Figure 10—Injection distribution into the 10 compartments for OH, ICD, and autonomous injector completions.

The pressure drops over the completion for the second permeability profile can be seen in Fig. 11. Pressure drop across the autonomous injection valve completion is around the constant limit of 30 bars, with lower differential pressure in the sections with lower permeability as these sections do not require to be restricted. Pressure drop across the ICDs is similar to the previous permeability profile, where it is lower than the autonomous injector while following the trend. The low permeability sections have moved due to the presence of fractures.

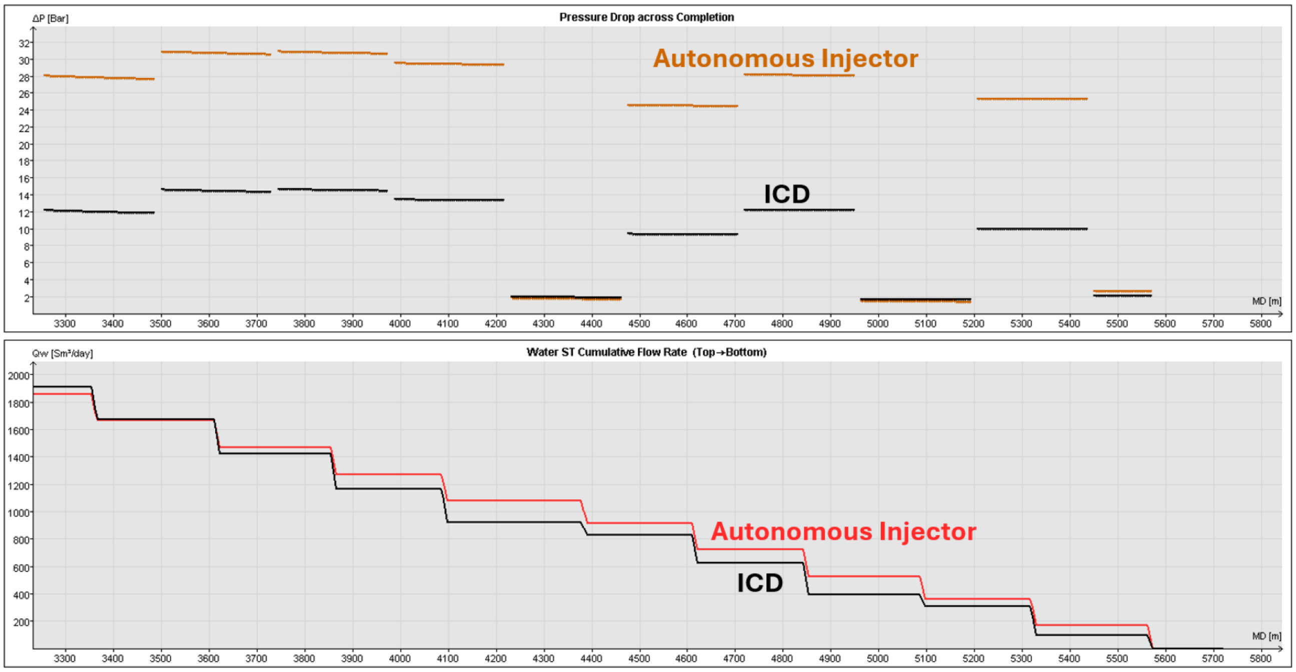


Figure 11—Downhole pressure profile and cumulative production for the fractured permeability profile. Top: Downhole differential pressure across autonomous injector and ICD completion. Bottom: Cumulative injection along completion for autonomous injector and ICD completions

Fig. 12 shows the distributed injection for each compartment. The sections with low permeability are now compartments 6, 3, and 1, where the OH and ICD completions are not able to inject water in the same capacity as the other compartments.

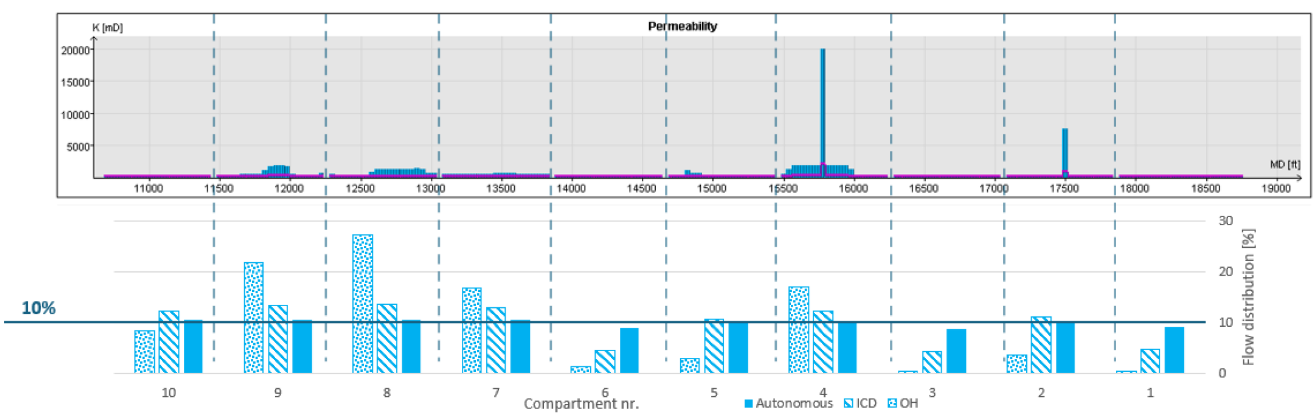


Figure 12—Injection distribution into the compartments for the second permeability profile

Dynamic reservoir simulation with autonomous injector

A simple box model with an injector and a producer well were simulated to look at the effect using the new valve instead of open hole injector completion. The producer is an open hole producer without any inflow control devices. Both wells are set up as multi segmented wells. The reservoir grid is 2000m × 2000m × 20m and is shown in Fig. 13. Two simulations cases have been set up for the dynamic simulation study:

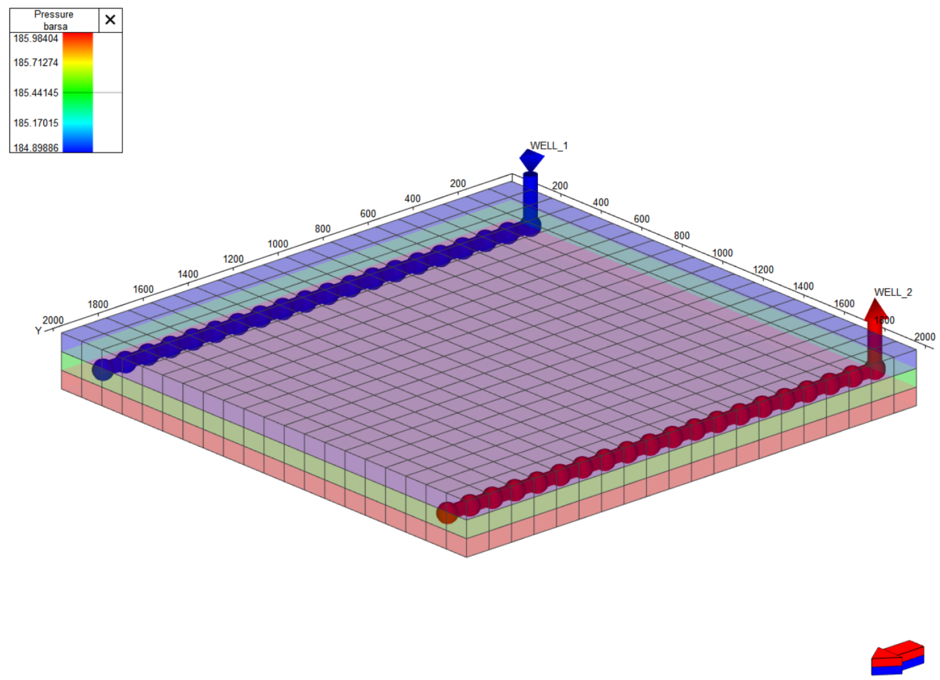


Figure 13—Modelled box reservoir showing the injection well (blue) and the producer well (red).

Case A: OH injector + OH producer

Case B: Autonomous injection valve + OH producer

The box model is separated into heterogeneous zones with different permeability in each zone. The permeability ranges from 10 mD to 200 mD in the most permeable zones. The purpose of this permeability distribution is to see how heterogeneous flow behavior will impact the effectiveness of the injector well, as some zones will enable higher flow of water than other zones. Fig. 14 shows the permeability profile for the reservoir, with the injector well on the left side and the producer well on the right side. In Case B the injector well has been segmented into 5 compartments with autonomous injector valves in each compartment, with packers placed in the low permeable section in between the producing zones.

The simulation was ran over a period of 3 years. The results can be seen in Fig. 15, with Case A as the striped lines and Case B with the autonomous injector as the solid line. As seen by these results there is a benefit of using autonomous valves for injection purposes. The well with autonomous valves delays the water breakthrough, which also delays the significant decline in oil rate that can be seen in the OH injection case. Cumulative production in Fig. 15b shows a very good oil gain for the autonomous valve, increasing the oil rate by approximately 8% while reducing the water produced by almost 50%. Snapshots in time are provided below in Fig. 16 to show the difference in water(blue)- and oil(red) saturation with and without completion with the new autonomous injection valve.

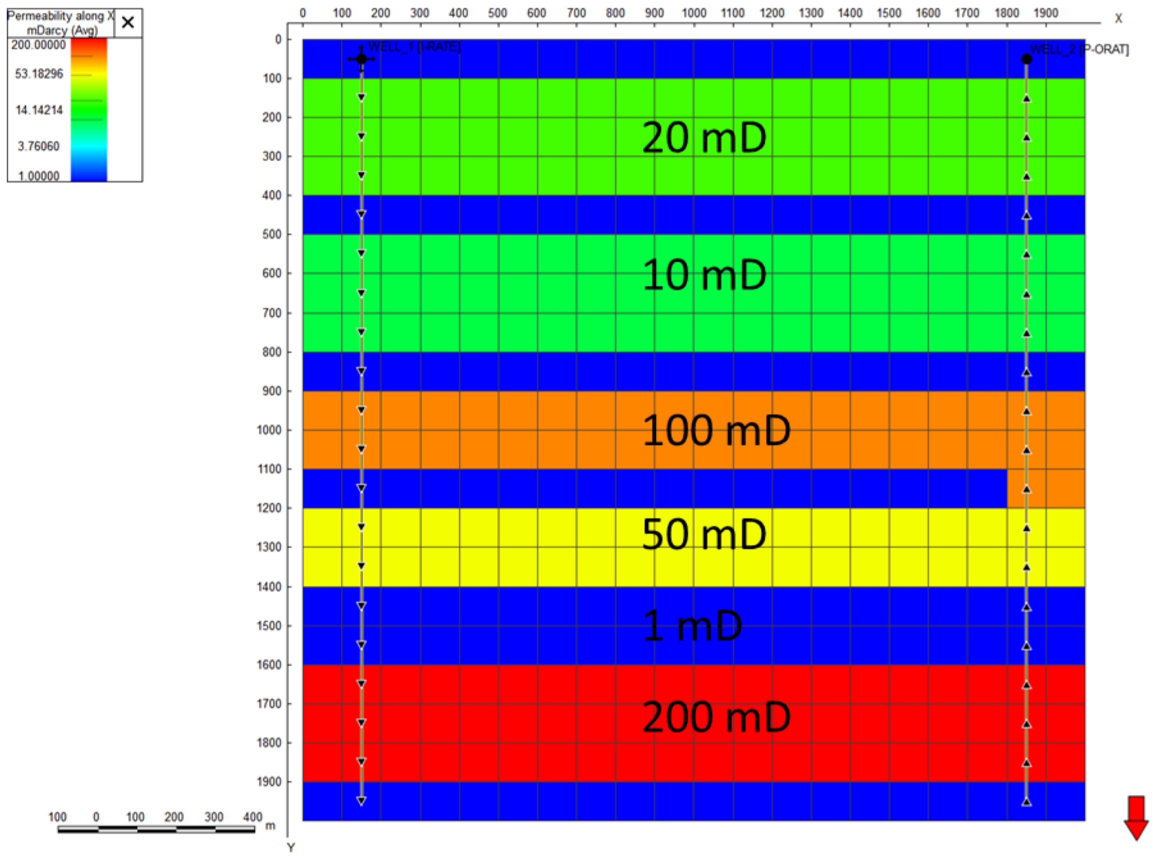


Figure 14—Permeability profile for the simulated reservoir. A heterogeneous reservoir is selected to show how the sweep in the reservoir is affected by completion technology.

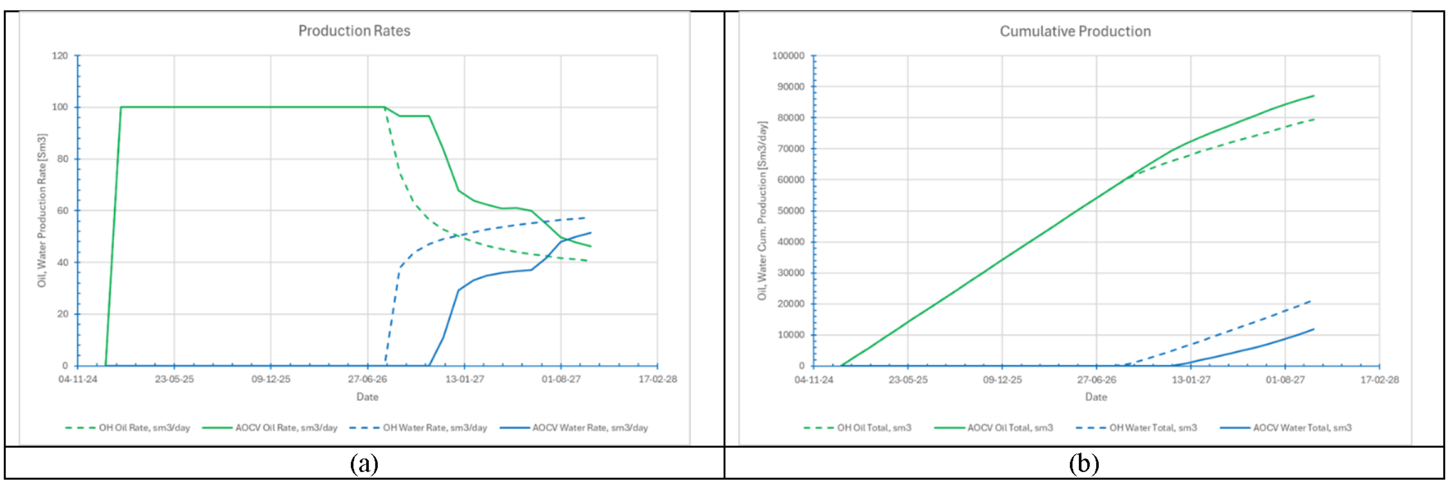


Figure 15—Results from simulation showing (a): production rates and (b): cumulative production with and without the autonomous injection valve installed on the injector side.

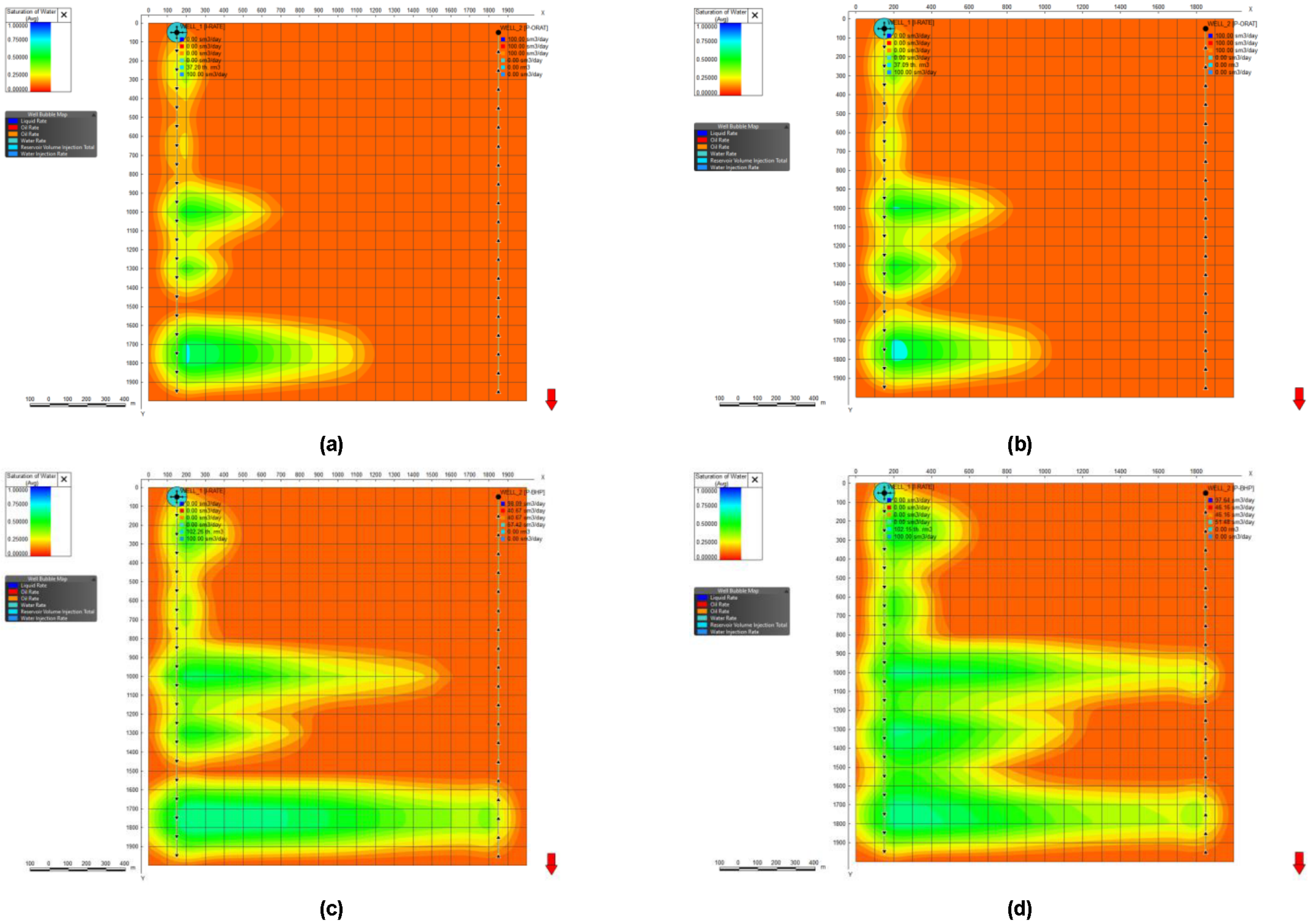


Figure 16—Water saturation plots for early life and late life. (a) Case A 01.01.2026, (b) Case B 01.01.2026, (c) Case A 01.10.2027, (d) 01.10.2027.

The saturation plots indicate that the initial differences between Case A and Case B are minimal. At later stages, the completion with the new valve demonstrates improved oil sweep across all permeability zones. From the saturation in Fig. 16c which shows the late timestep for case A most of the injected water flows in the high permeable zones 3 and 5, causing oil to bypass in the remaining zones. Looking at the saturation in Fig. 16d which shows late life saturation for Case B, the water has provided a better sweep in the lower permeability sections, most noticeably in zones 3 and 4, while still sweeping the high permeable zone 5. There is also better sweep in the initial zones 1 and 2, although not as noticeable as the other zones. Overall, the autonomous completion is improving the sweep and reducing the amount of bypassed oil that will remain untouched by the injected water, which compliments the cumulative results from Fig. 15b.

Discussion and further work

The presented valve is proven to provide a constant flow rate over a range of differential pressures. It shows ability to withstand harsh environmental conditions with solid particles and well fluids. The work done on generating a mathematical representation of the valve has proven to be valuable. Still, some improvements to this approach have been identified, such as more advanced friction modelling and to include manufacturing tolerances.

Simulation results demonstrate that implementing the autonomous injection valve has a potential to increase oil production, reduce water production, and provide more efficient reservoir sweep over time. Water saturation plots reveal limited differences between cases at early stages; however, the valve's effectiveness in improving oil displacement across heterogeneous zones becomes more evident in late life. The ability of this technology to adapt injection profiles enhances recovery and mitigates early water breakthrough.

It is important to note that these findings are based on idealized models and may not reflect all operational complexities. More advanced reservoir simulations together with other autonomous inflow control technologies on the producer side must be explored. Pilot testing and field validation are required to confirm simulation outcomes and address any potential risks.

Conclusions

- A new autonomous injection valve has been developed and proven to work as intended.
- A mathematical model was developed to output the valve internal geometry with the goal flow rate as input.
- The valve has proved to be resilient to harsh conditions with particles and well fluids.
- Simulation of the valve in a simple model shows its potential to tackle a large problem with injector wells.'
- The autonomous injection valve has been implemented into both static and dynamic modelling tools with success.
- Modelling results show how the valve restricts flow to thief zones, providing even injection, even in fractured reservoirs.
- Homogenous injection provides increased oil for producer wells in dynamic modelling.

Nomenclature

ppm	Parts per million,concentration of particles (weight based)
OH	Open hole (completion)
mD	milli-Darcy (permeability)
ICD	Inflow control device
DP	Differential pressure

References

1. C. Nomme, "A fluid flow control device and method," WO Patent 2025125414 A1, Jun. 19, 2025.